

GY457

Applied Urban and Real Estate Economics



Topic 8:

Housing Issues in China: Facts, Causes, and Policies



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Housing Issues in China: Facts, Causes, and Policies



Introduction

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 - BSc (Tongji, Shanghai) → Exchange (St.Gallen, Switzerland) → MSc REEF (LSE) → PhD (LSE)
 - Research: Urban Economics, Real Estate, Policy Evaluation, Applied Microeconomics (with a focus on China and the UK)
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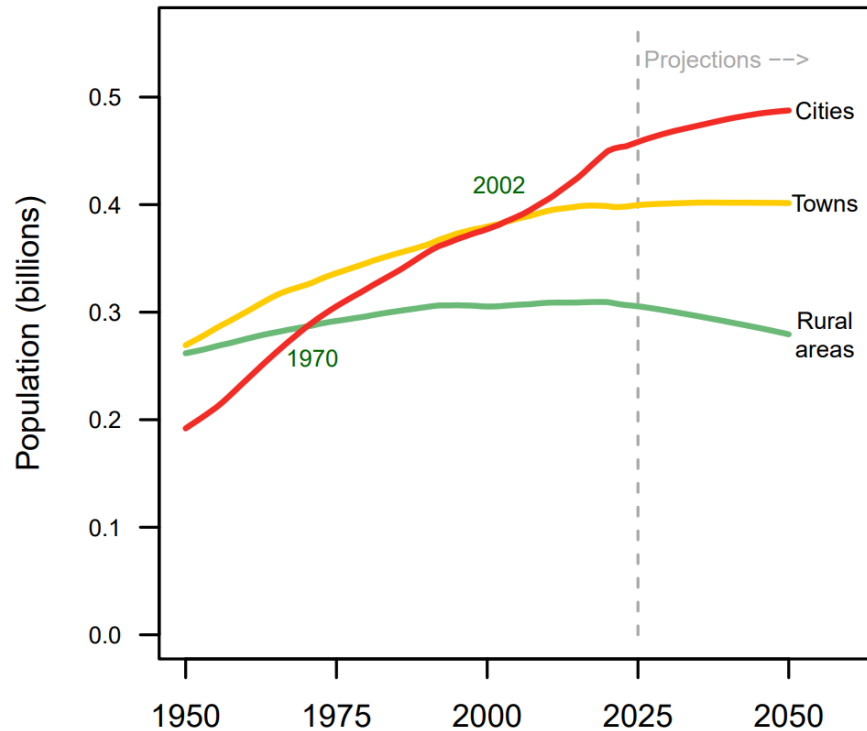
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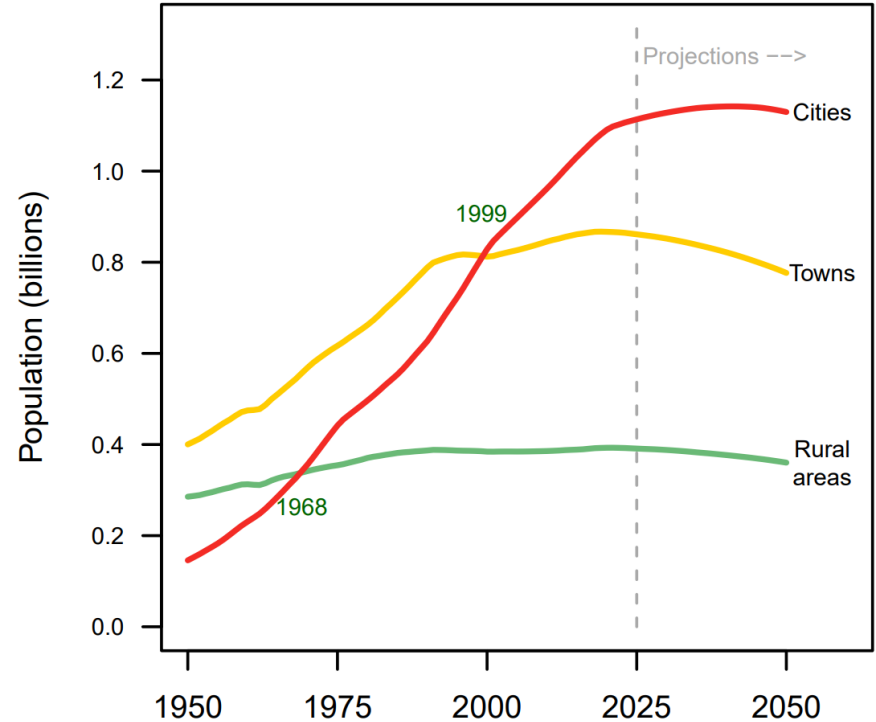
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Stylized fact #1: Asia and particularly China have been urbanizing at a breath-taking pace

Europe, Northern America, Australia and New Zealand



Eastern and South-Eastern Asia, and Oceania (excluding Australia and New Zealand)

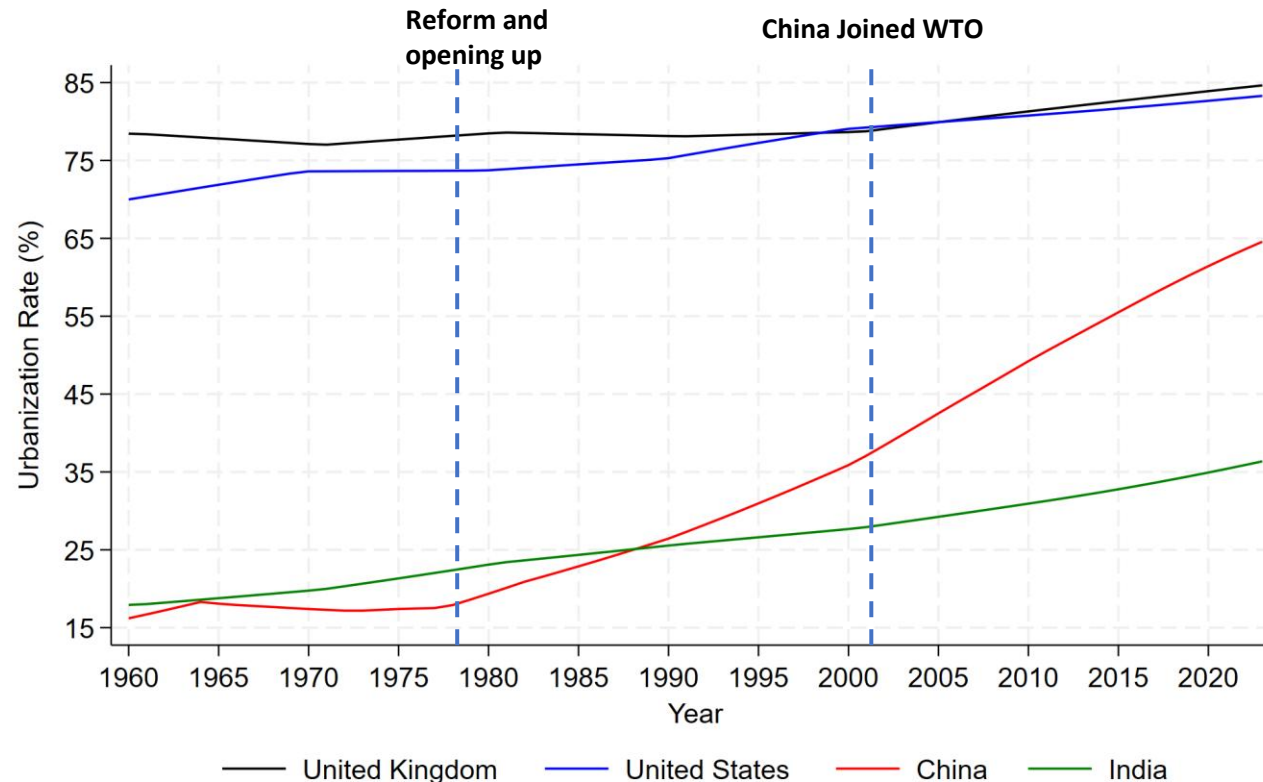


Note: “Cities”, “towns” and “rural areas” refer to standardized categories based on population size and density

Source: World Urbanization Prospects 2025 (United Nations, 2025)

Stylized fact #1: Asia and particularly China have been urbanizing at a breath-taking pace

- Urbanization rates (share population living in urbanized areas)



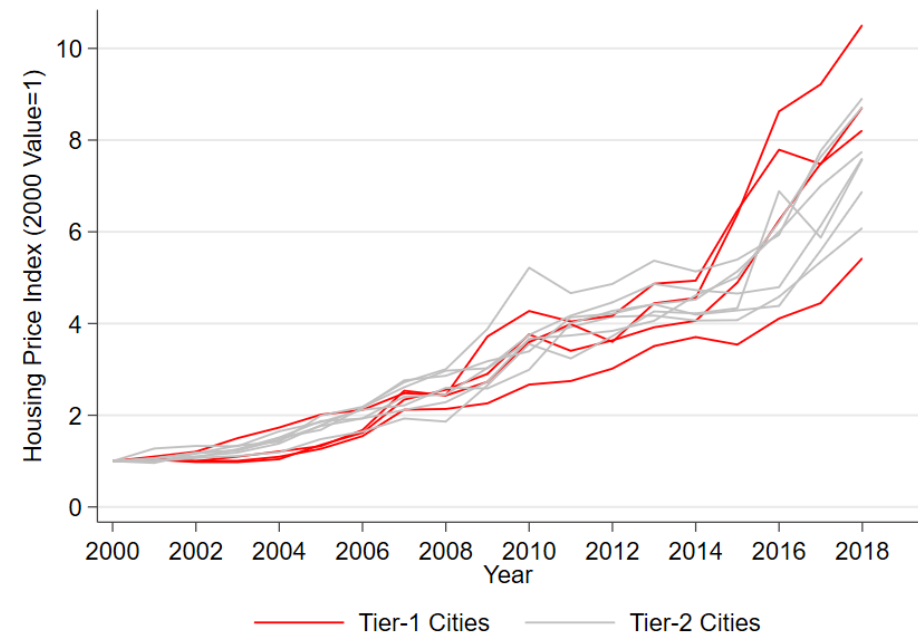
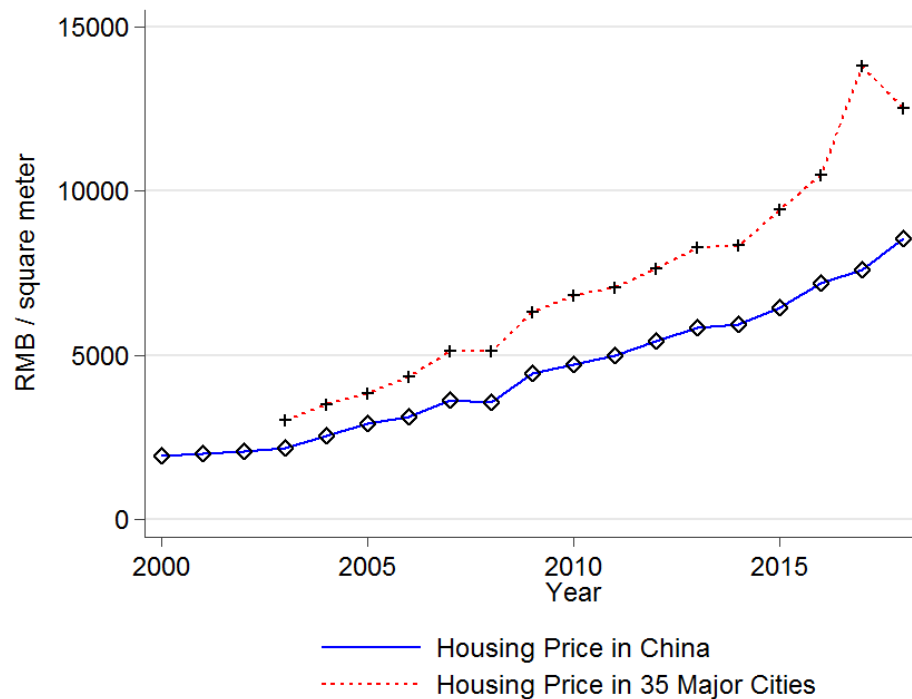
Source: World Bank



China has been urbanizing rapidly since at least 1950 (with exception of Cultural Revolution years) but now slowing down. The urbanization rate in China is 67% in 2024 (National Bureau of Statistics)

Stylized fact #2: Urbanization is associated with urban housing affordability crisis

- Before COVID: Avg. nominal HP increases by 313% (annual growth rate 8%) in 35 major cities in China from 2003 to 2018
 - Nominal HPs in **Tier-1** cities increase even faster

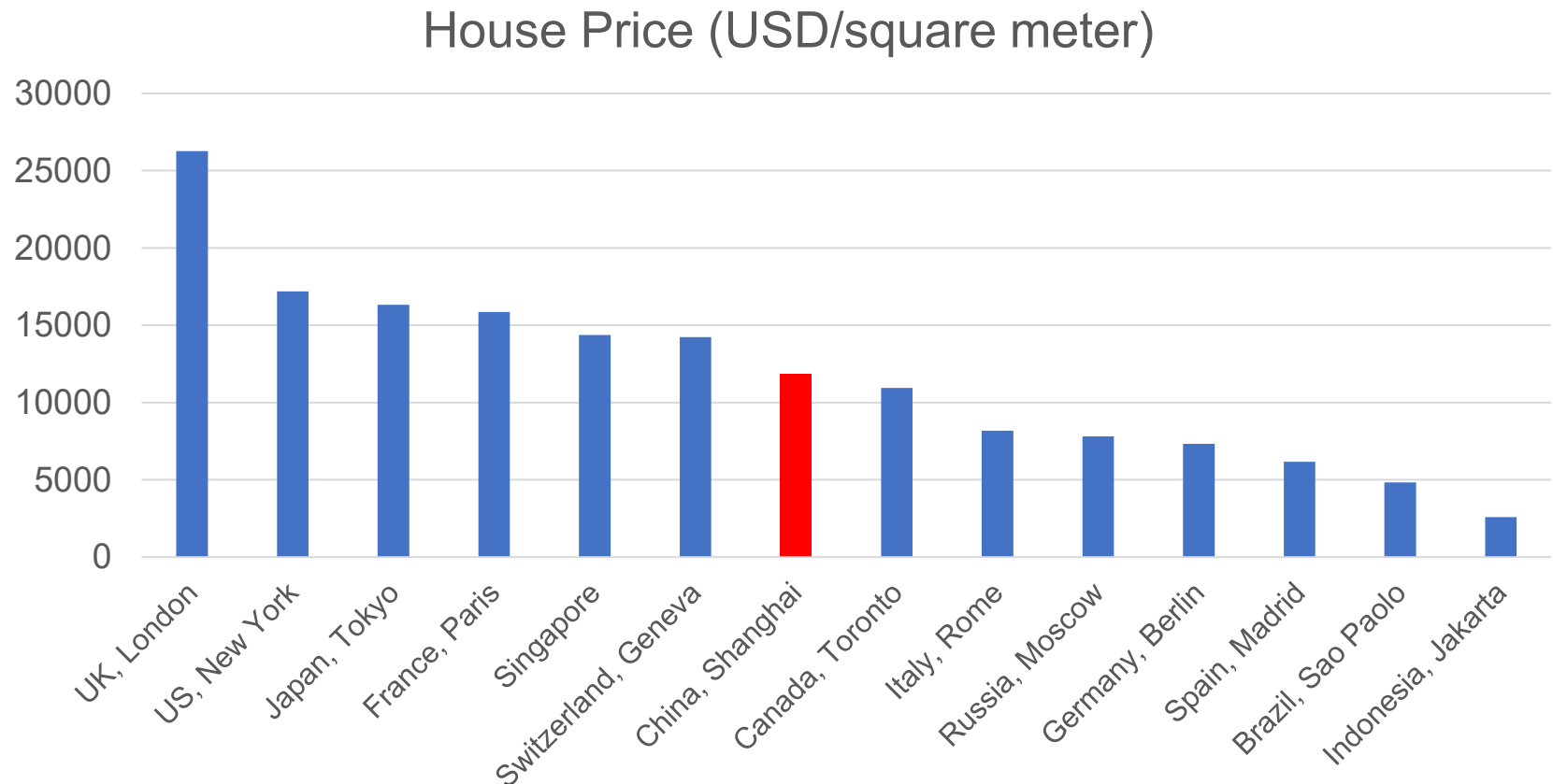


1 RMB = 0.11 GBP (27 Sep 2019, 21:00 UTC)

Source: National Bureau of Statistics, city-level and province-level statistical yearbooks & self calculation

Stylized fact #2: Urbanization is associated with urban housing affordability crisis

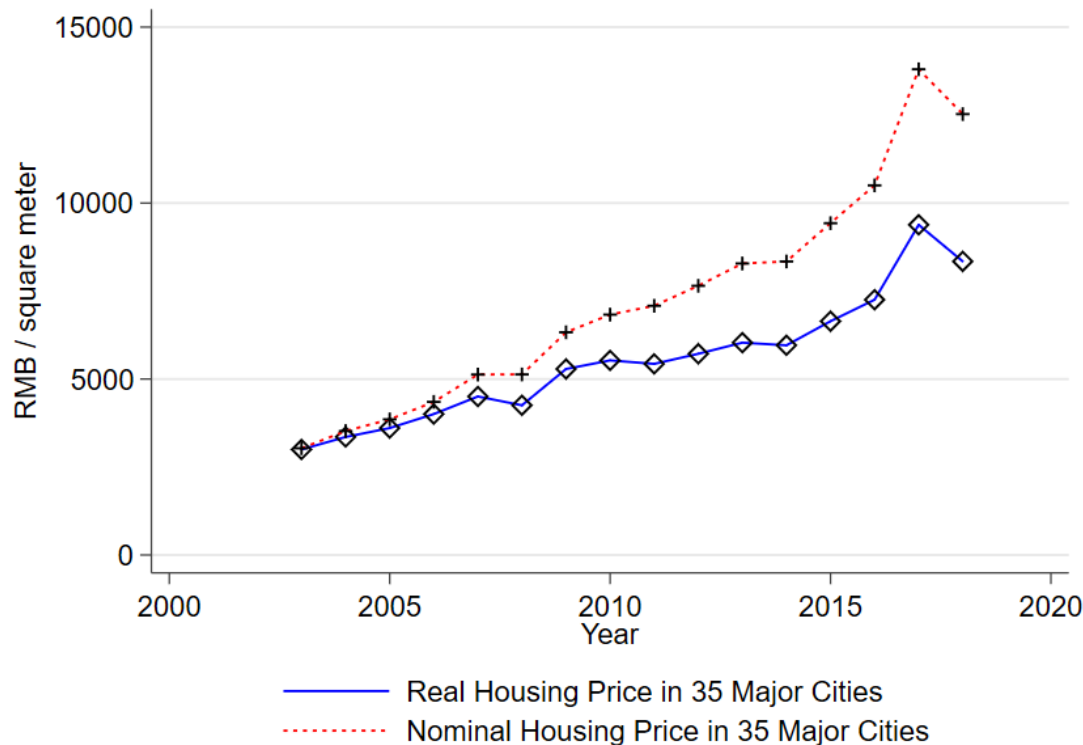
- HPs (in 2019 US\$) of big cities in China are comparable with HPs in superstar cities in developed countries



Source: Global Property Guide
Last accessed: 10 Oct 2019

But what about *real* house prices?

- Nominal HPs represent the monetary values of housing. Real HP is the nominal HP adjusted for inflation
- Avg. real HPs increase by 178% (ann. growth rate of 4% real vs. 8% nominal) in 35 major cities (2003-2018, 2003 RMB)

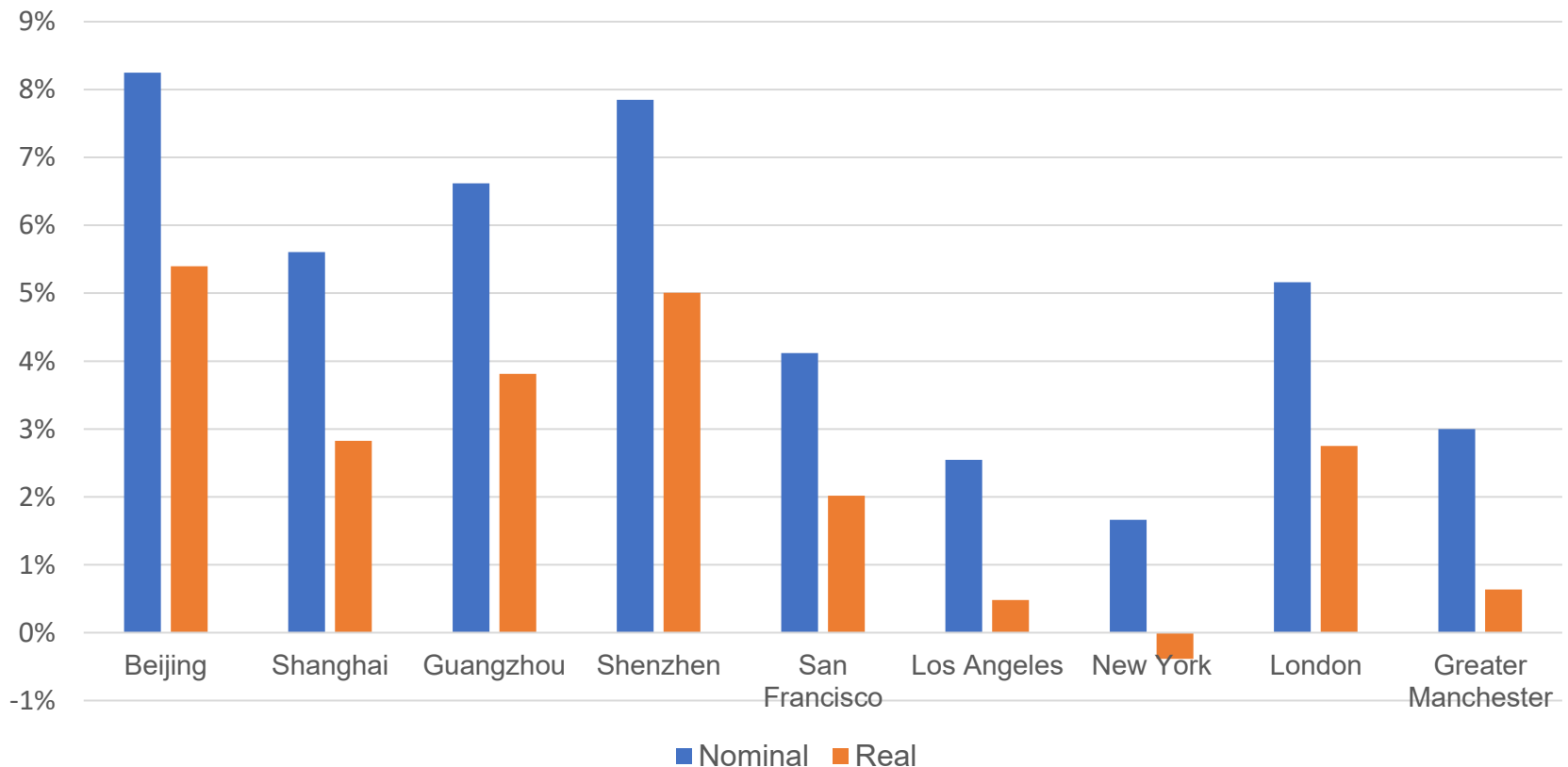


Source: National Bureau of Statistics & self calculation

Stylized fact #2: Urbanization is associated with urban housing affordability crisis

- Between 2005 and 2019, HP growth rate is relatively high in China compared with HP growth rate in the US and UK

Nominal and Real HP Growth Rate Between 2005 and 2019



Source: Land Registry, Federal Housing Finance Agency, National Bureau of Statistics & self calculation

Excursus: Chinese city tier system

- The Chinese city tier system is a hierarchical classification of Chinese cities. There are no such official lists in China
- However, it is frequently referred to by various media publications. Cities in different tiers reflect differences in consumer behavior, income level, population size, business opportunity, etc.

Tier	Cities	Number of cities
Tier 1	Beijing, Shanghai, Guangzhou, Shenzhen	4
New Tier 1	Chengdu, Hangzhou, Wuhan, Chongqing, etc.	15
Tier 2	Xiamen, Fuzhou, Wuxi, Hefei, etc.	30
Tier 3	Weifang, Baoding, Zhenjiang, Yangzhou, etc.	70

Source: Yicai Global 2017

Chinese city tier system

- Tier-1 city: Shanghai

- Tier-2 city: Hefei

- Tier-3 city: Suqian



Population (Million)	24.2
Disposable Income (RMB)	57,692
GDP (Billion RMB)	2,818
GDP per capita (RMB)	116,562
House price (RMB/m ²)	25,910

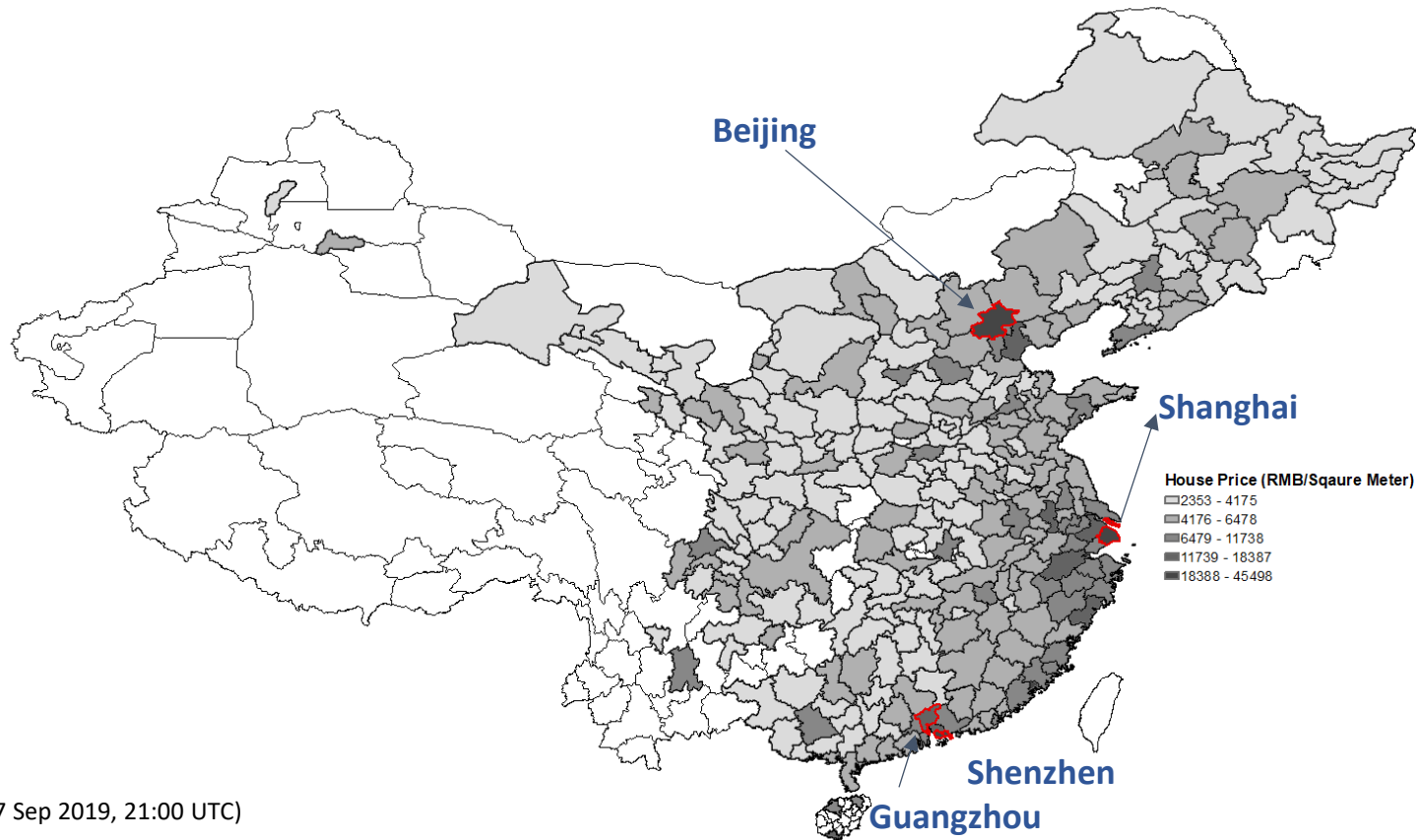
Population (Million)	7.9
Disposable Income (RMB)	34,852
GDP (Billion RMB)	627
GDP per capita (RMB)	80,138
House price (RMB/m ²)	9,312

Population (Million)	4.9
Disposable Income (RMB)	24,086
GDP (Billion RMB)	235
GDP per capita (RMB)	48,311
House price (RMB/m ²)	4,514

Note: data in 2016, 1 RMB = 0.11 GBP (27 Sep 2019, 21:00 UTC).

Stylized fact #2: Urbanization is associated with urban housing affordability issues

- In consistent with economic developments, HPs in Tier-1, Tier-2 cities and cities along the southeast coast are higher



Notes:

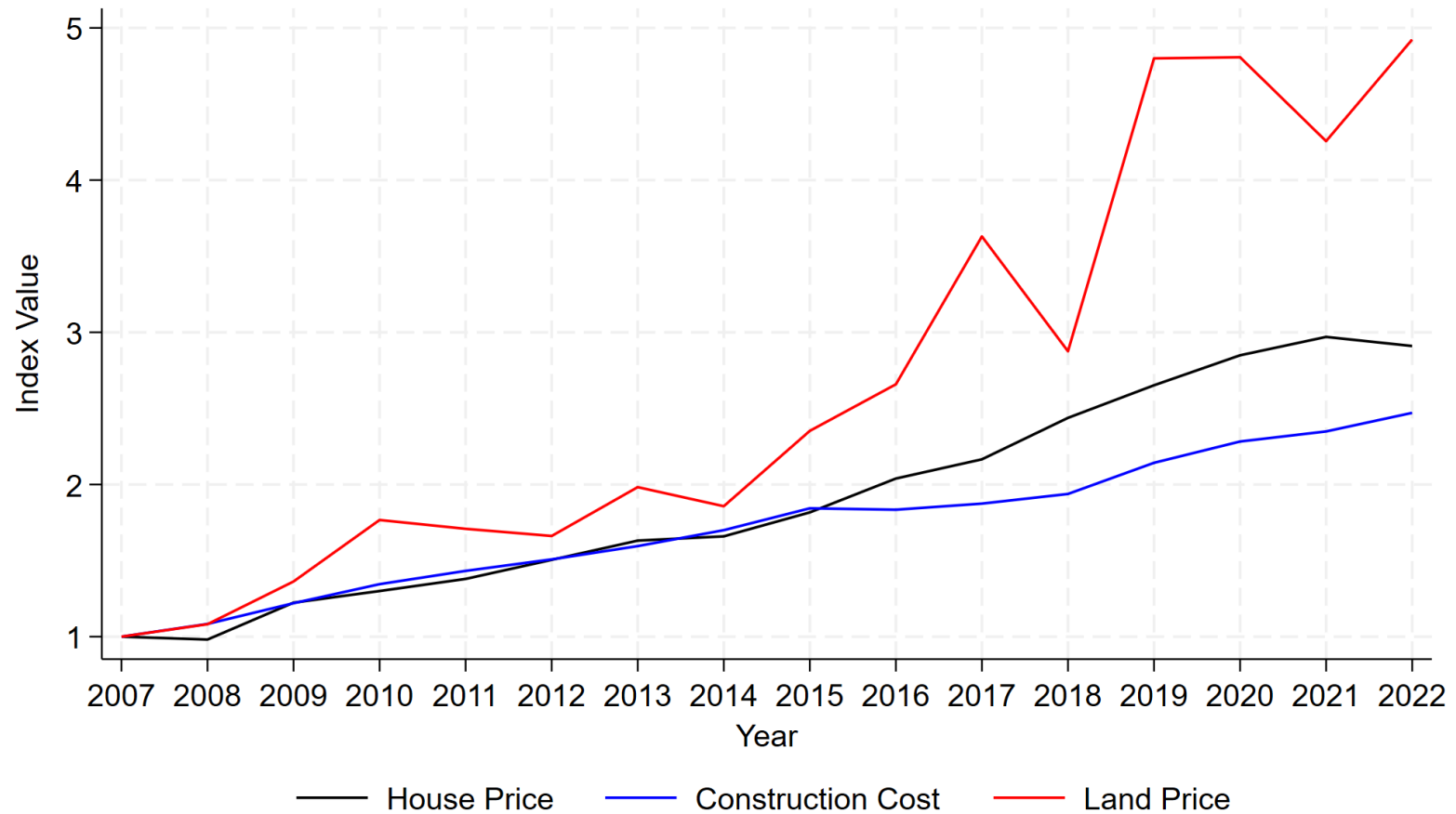
1 RMB = 0.11 GBP (27 Sep 2019, 21:00 UTC)

HP data in 2016

A 100 square meter house in Beijing in 2018: 3,742,000 RMB (411,620 GBP)

Stylized fact #3: Strong HP growth is mainly driven by land prices and not by construction costs

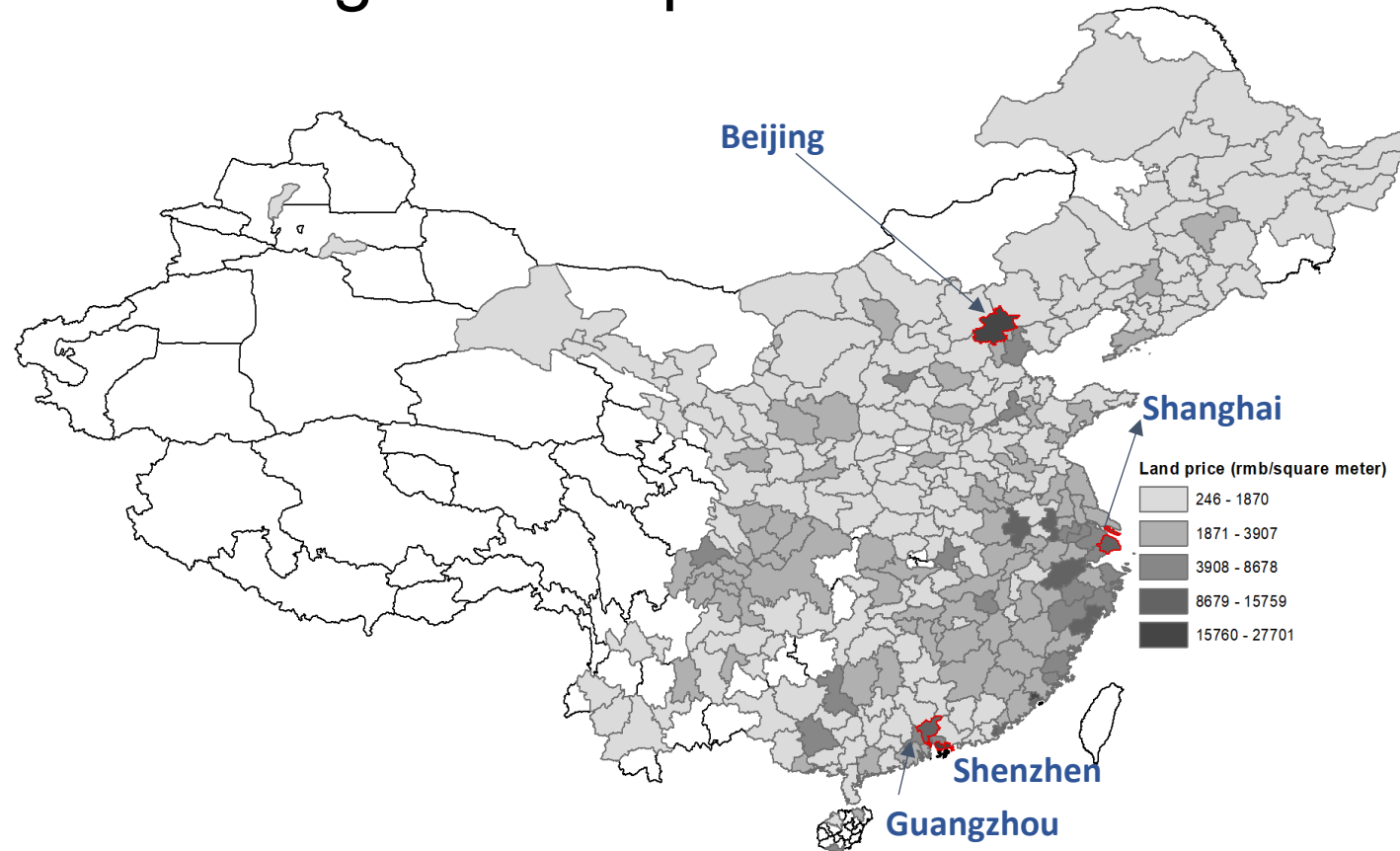
- Land prices increase faster than construction costs



— House Price — Construction Cost — Land Price

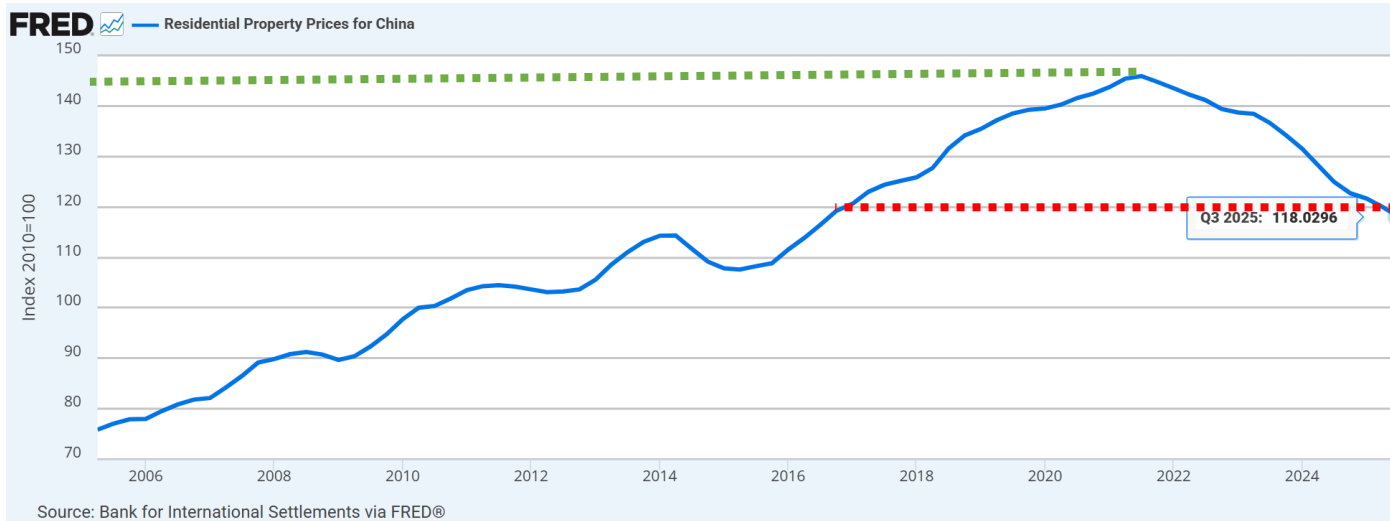
Stylized fact #3: Strong HP growth is mainly driven by land prices and not by construction costs

- Tier-1, Tier-2 cities, and cities along the southeast coast have higher land prices

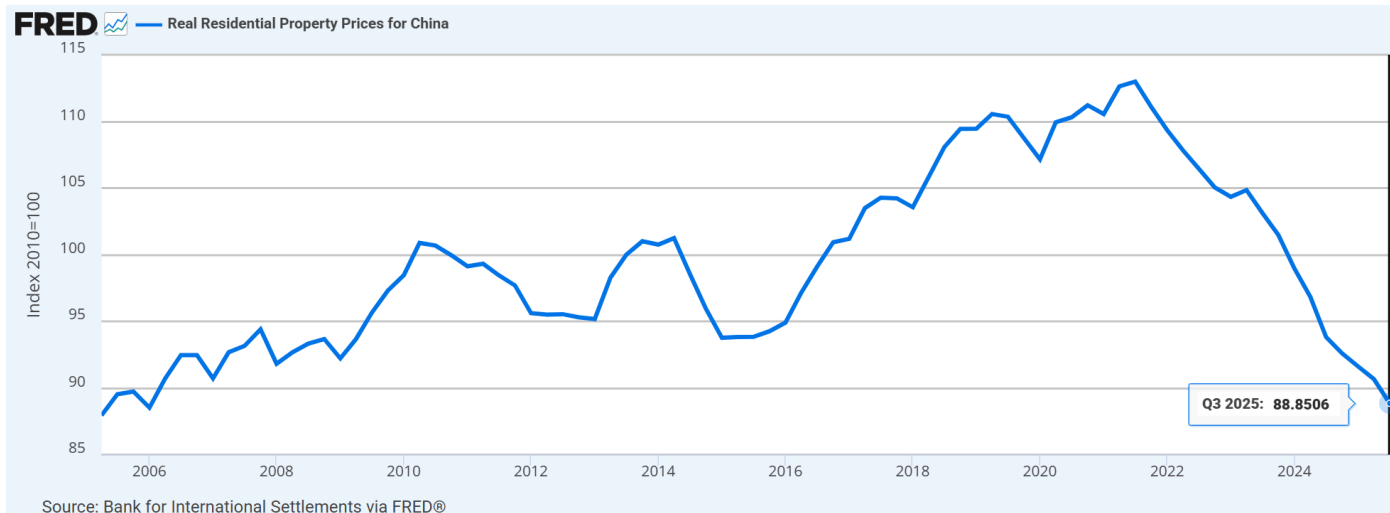


Note: 1 RMB = 0.11 GBP (27 Sep 2019, 21:00 UTC)
Land price data in 2016

Stylized fact #4: HPs decline after COVID



24% nominal HP decrease and 27% real HP decrease in Q3 2025 compared with Q3 2021

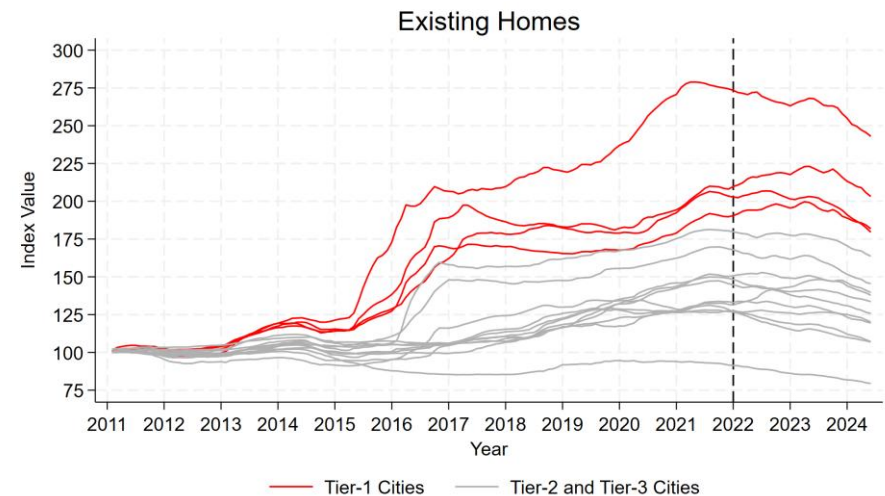
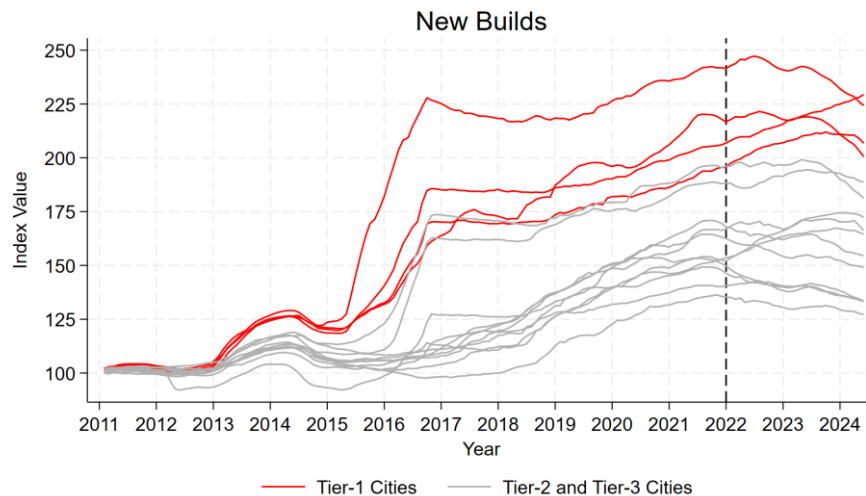


Sources: <https://fred.stlouisfed.org/series/QCNN628BIS#>
<https://fred.stlouisfed.org/series/QCNR628BIS#>

Coverage includes Q1 2007-Q4 2015: New dwellings in 70 cities in China; from Q1 2016: Existing buildings in 70 cities in China. The real HP series is deflated using CPI.

Stylized fact #4: Spatial variations in HPs after COVID

- Based on the 70 large and medium-sized cities HP index:
 - 0.5% HP **increase** (new builds) and 8.1% HP **decrease** (existing homes) in Tier 1 cities between September 2021 and May 2024
 - 6.6% HP **decrease** (new builds) and 12.3% HP **decrease** (existing homes) in lower Tier cities between September 2021 and May 2024



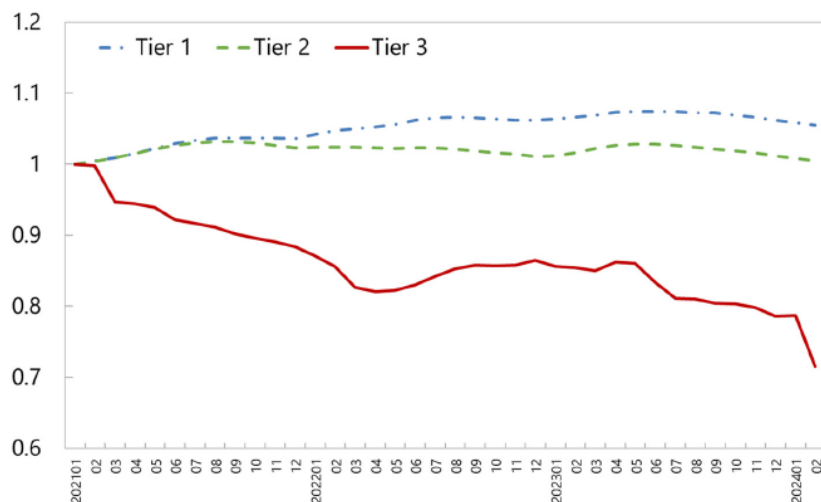
Source: National Bureau of Statistics & self calculation

Stylized fact #4: Spatial variations in HPs after COVID

- House prices have levelled off in Tiers 1 and 2 cities, but have fallen sharply in Tier-3 cities
- Real estate is running into diminishing returns to growth, mainly driven by Tier-3 cities

House Price Changes by City Tier (2021:01 to 2024:02)

(2021:01=1)



Sources: NBS and author calculations. See Appendix I.3.

Fig. 3. House Prices Change by City Tier (2021:01 to 2024:02).

Real Estate and Growth.

Variables	Real GDP growth
Instrumented real estate investment to GDP ratio	0.492*** (0.064)
Instrumented real estate investment to GDP ratio × Period 09–15	–0.148** (0.068)
Instrumented real estate investment to GDP ratio × Period 16–19	0.045 (0.082)
Instrumented real estate investment to GDP ratio × Period 20–22	–0.596*** (0.085)
Real GDP per capita	–0.078*** (0.007)
Population growth rate	0.014 (0.035)
Urbanization rate	0.002*** (0.000)
Constant	0.630*** (0.054)
Number of observations	5689
R squared	0.341
Year fixed effect	YES
City fixed effect	YES

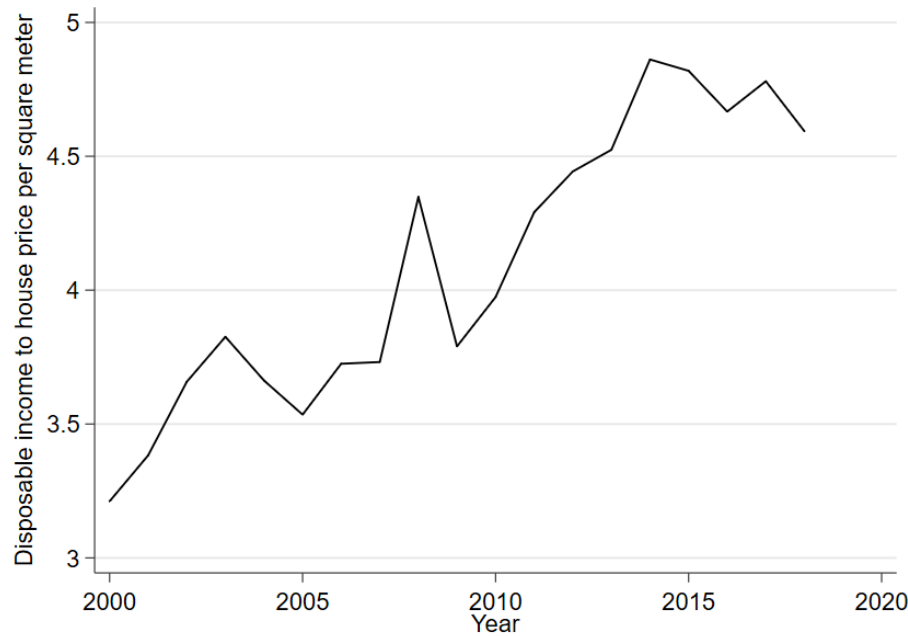
Source: Rogoff and Yang (2024)

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Housing affordability

- Is Chinese housing boom driven by fundamentals? The ratio of disposable income relative to housing price per square meter is generally increasing over the past 20 years
- However, housing is very unaffordable in Tier-1 cities



Source: National Bureau of Statistics & self calculation

Housing affordability: international comparison

- Demographia uses *median housing price* relative to *median household income* to measure housing affordability and make international comparison

Nation	Affordable (3.0 & Under)	Moderately Unaffordable (3.1-4.0)	Seriously Unaffordable (4.1-5.0)	Severely Unaffordable (5.1 & Over)	Total	Median Market
Australia	1	1	5	16	23	5.7
Canada	12	16	5	17	50	4.0
China (Hong Kong)	0	0	0	1	1	20.9
Ireland	2	2	1	0	5	3.7
New Zealand	0	0	2	6	8	6.5
Singapore	0	0	1	0	1	4.6
United Kingdom	0	4	18	11	33	4.8
United States	56	70	34	28	188	3.5
TOTAL	71	93	66	79	309	4.0

Source: 15th Annual Demographia International Housing Affordability Survey: 2019

House price to income ratio in China

- Assumptions:
 - 2.62 person per household based on 2020 Census data
 - Average floor area per person = 36.52 m² based on 2020 Census data

$$\text{Housing affordability} = \frac{\text{Housing Price per m}^2 \times \text{Floor Area Per Person} \times \text{Household Size}}{\text{Average Salary} \times \text{Household Size}}$$

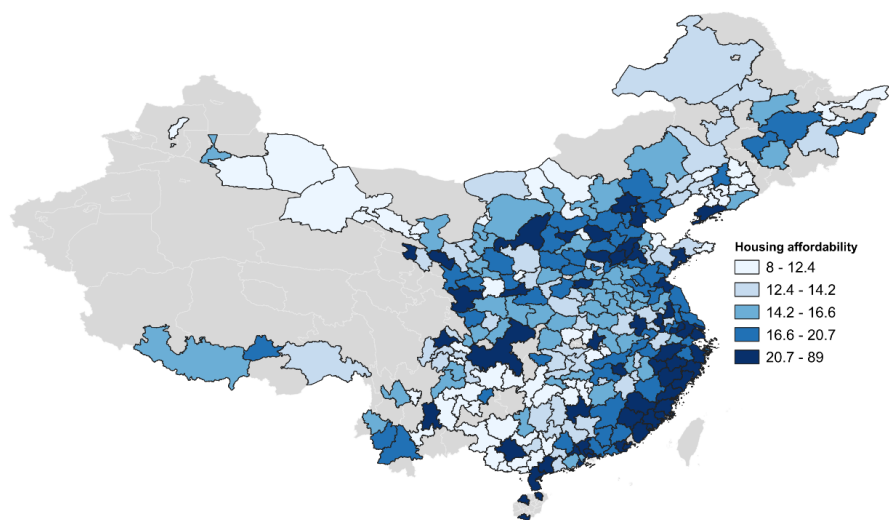
Tier-1 Cities	Ratio	Tier-2 Cities	Ratio
Beijing	21	Tianjin	11.6
Shanghai	18.2	Xiamen	17.1
Guangzhou	15	Nanjing	13.9
Shenzhen	31.8	Hangzhou	14.6
San Francisco	8.8	Wuhan	10.2
London	8.3	Chengdu	10.2

Source: 15th Annual Demographia International Housing Affordability Survey: 2019, National Bureau of Statistics, and China Statistical Yearbook. City level data in 2021 & self calculation. If household employment rate is not 100%, the ratio will be higher.

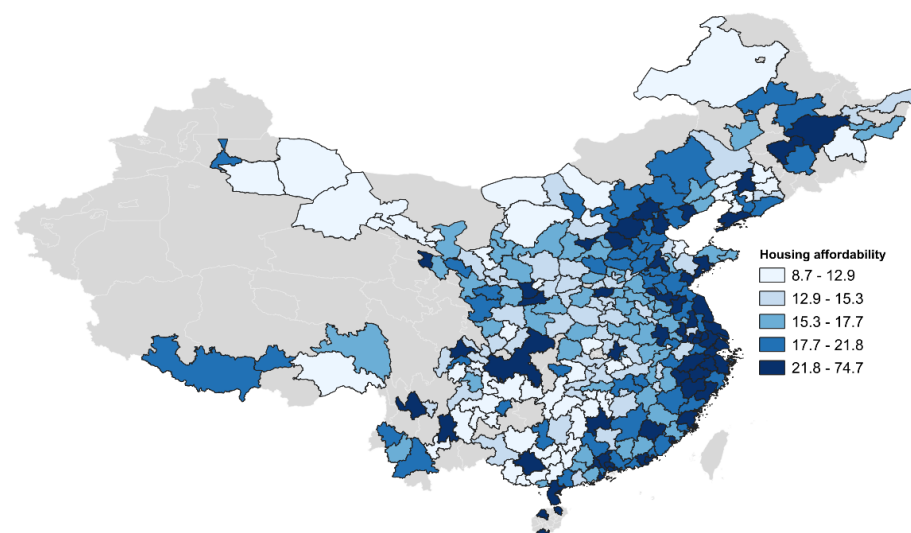
House price to income ratio in China — *cont.*

- Based on Demographia's criteria, housings are “severely unaffordable” in many Chinese cities
- There are also spatial variations in housing affordability status

Existing Homes



New Builds



Note: Both maps present the house price (defined as 100 square meters multiplied by the average house price per square) divided by the disposable income ratio in 2024

Source: Yu (2026)

Other measures of affordability in China

- Li, Qin and Wu (2020) measure housing affordability in China from both the macro and micro perspectives:
 - Homeowners: house price & repayment to income ratio
 - Renters: rent to income ratio
 - Panel data of city-level housing affordability: available online
- They find that the overall housing affordability in Chinese cities remained relatively stable or even improved between 2014 and 2018
- A few “superstar cities” are associated with serious housing affordability problem

Housing market related concerns in China

- Real HPs increase by 178% in 35 major cities in China b/w 2003-2018:
 - In US before Great Financial Crisis b/w 1991-2006: about 50%
 - In UK before Great Financial Crisis b/w 1991-2006: about 124%
 - House price to income ratio is extremely high in Tier-1 and most Tier-2 cities in China
 - Yet: High vacancy rates and over-supply of housing in some cities...
- ⇒ Is there a 'housing bubble' (irrational exuberance) in China? Or are there other explanations?

Discuss...

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Overview of housing policies

1. Purchase restriction policy

- Minimum down payment requirements
- Only households with local *hukou* (household registration) or who can prove they have worked locally for a consecutive number of years are eligible to purchase homes

2. Property tax

- No nationwide property tax
- Pilot programs in two cities since 2011: 0.6% or 0.4% in Shanghai; 0.5% to 1.2% in Chongqing. Both cities introduced exemption rules
- Bai et al. (2015) find that property tax lowered the Shanghai average HP by 11%–15% but raised the Chongqing average HP by 10%–12% (driven by a potential spillover effect from high-end to low-end properties)
- 2.73% more decrease in HPs for higher-taxed houses in Shanghai (Lyu, 2024)

3. Sales restriction policy

- LGs prohibit re-sale of properties for a consecutive number of years (to limit speculative behavior)
- LGs impose an upper-limit sale price for certain new builds/developments

4. Afford. housing scheme & shared ownership scheme

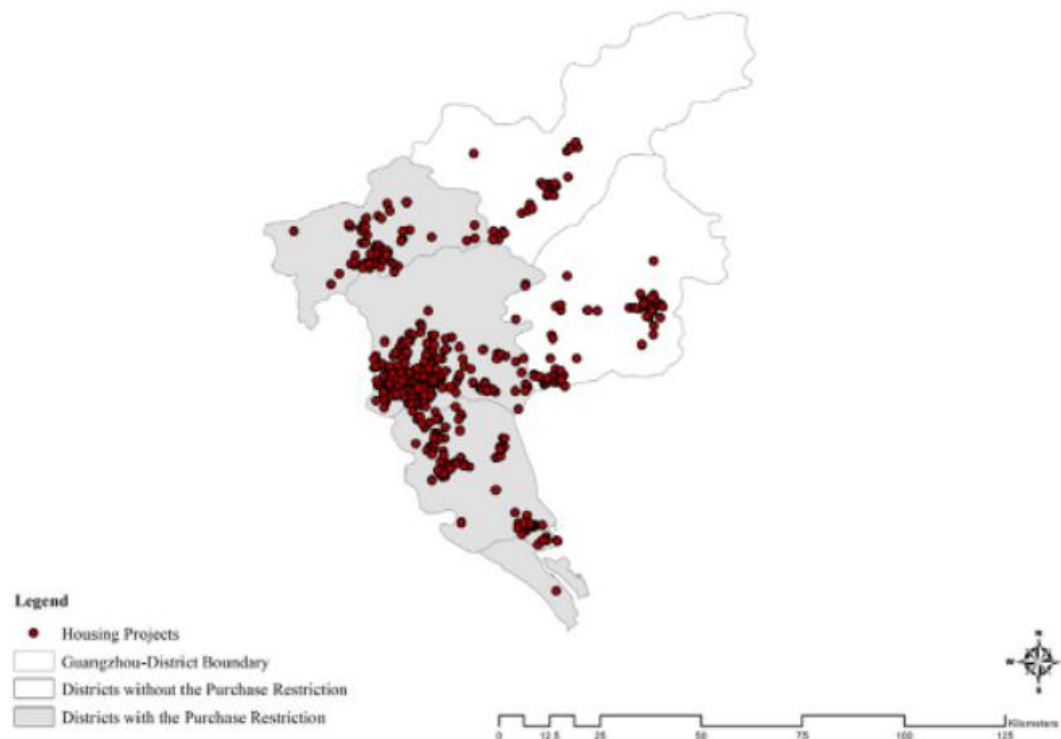
Policy evaluation

- Somerville *et al.* (2020) study the effect of home purchase restriction policy
 - Exploit variation in purchase restrictions within a city by district limits
- Data
 - Transaction prices & number of transactions (for newly constructed condominiums) come from China RE Information System (CREIS)
 - 2,014 development projects in Chengdu, Guangzhou, Hefei, and Qingdao
- Method: Diff-in-Diff

Policy evaluation

- The DiD estimation compares
 - HPs and number of transactions within a city by district
 - HPs close to the district border

Purchase Restriction by Districts and Housing Projects in Guangzhou



Policy evaluation

- Key findings

1. In short run, restrictions on non-owner-occupant purchases significantly reduce purchase-activity levels by 40%
 - However, effects diminish over time
2. There is no relative change in either the number of land auctions or prices paid for land
3. There are no relative changes post-policy introduction in house prices b/w restricted & unrestricted areas (contradicting Du & Zhang 2015): downward sticky price response (loss aversion)
4. While buyers were affected by the restrictions, developers behaved in a manner consistent with an expectation that the restrictions were temporary and that waiting to sell would be more profitable than dropping prices

Discussions

1. Purchase restriction policy

- Could it address the affordability problem in the long term? Somerville et al. (2020) show that the restriction leads to fewer transactions but has an insignificant effect on house & land prices
- Will young first-time buyers find it more difficult to afford homes due to the credit constraint? Will this policy discourage internal migration?

2. Property tax

- The design of the property tax: second and above homes? high-end homes (spillover)? tax rates? redistributed? selective?
- Will property taxes affect local housing markets adversely and influence local fiscal conditions?

3. Affordable housing scheme

- Who pays for the subsidy: local governments? developers?
- How to design the allocation system?

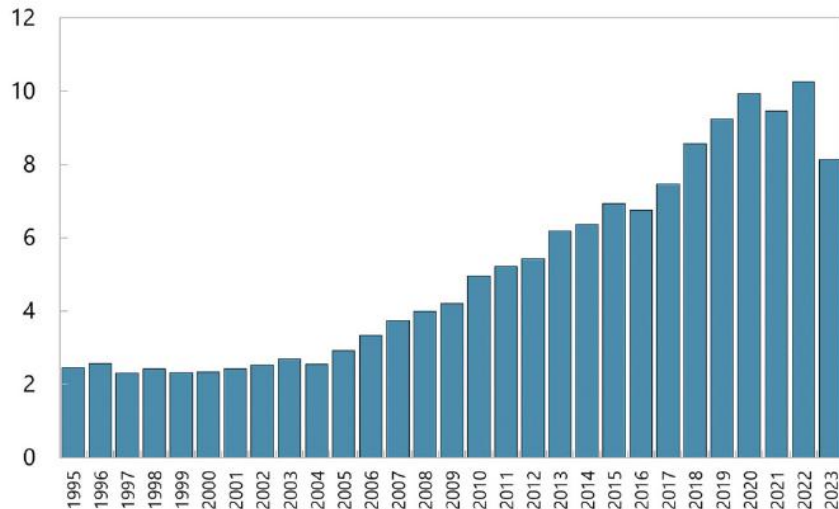
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Overbuilding?

- “Housing under construction over housing completed” indicates the number of years required to finish existing housing projects at the current construction rate
- A typical project requires 1-3 years for completion, and a high ratio can be expected in a rapidly growing market. However, ratios exceeding 10 might suggest a market in distress where developers struggle to finish projects because of a shortage of home purchasers and financing

Housing Under Construction Over Housing Completed

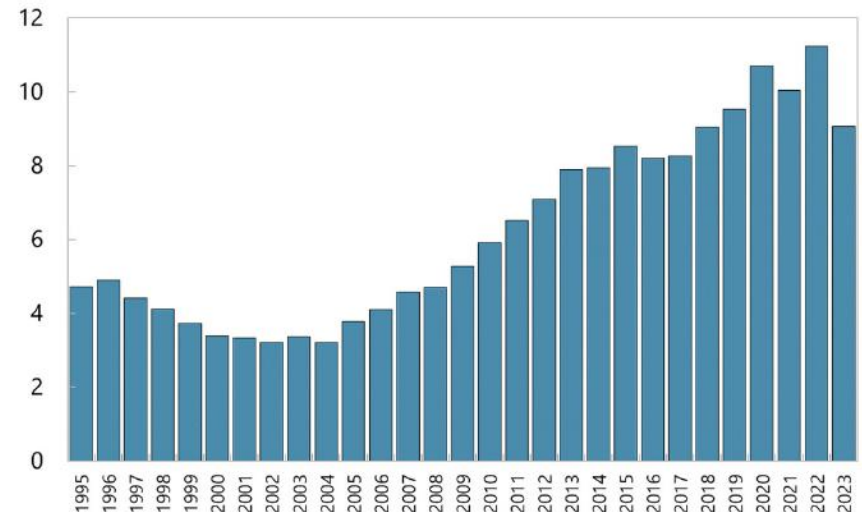


Sources: NBS and author calculations

Fig. I.6. Housing Under Construction Over Housing Completed

Source: Rogoff and Yang (2024)

CRE Under Construction Over CRE Completed



Sources: NBS and author calculations

Fig. I.7. CRE Under Construction Over CRE Completed

“Ghost towns”

- between 2000 and 2011, newly constructed areas has increased by 76.4%, whereas the size of urban population has expanded by 50.5%... Motivated by land sale revenue, LGs prefer “constructing cities” (Wang et al., 2015)



An empty street in front of a vacant residential complex in Kangbashi, Ordos

Source: <https://www.businessinsider.com/chinas-ghost-cities-in-2014-2014-6?r=US&IR=T>

“Ghost towns”

- “There were roughly 63,000 people living in Kangbashi in 2013... While this district was originally planned for 300,000 people and most of the homes were still empty... in 2017, there are now 153,000 people living there, and housing prices have risen roughly 50% on average from the end of 2015. Of the 40,000 apartments that had been built in the new district since 2004, only 500 are still on the market” (Forbes, 2017): Planning and building ahead of urbanization?
- However, a joint study by Peking University and Baidu in 2015 still found high vacancy rates in parts of Kangbashi

Sources:

<https://qz.com/540571/baidu-found-chinas-ghost-cities-but-it-is-keeping-their-locations-mostly-a-secret/>

<https://www.forbes.com/sites/wadeshepard/2017/06/30/ordos-chinas-most-infamous-ex-ghost-city-continues-rising/?sh=743f5c7e6877#6aff89a46877/2017/06/30/>

Vacancy rate

- Since 1985 no more *official* estimates available, but data is available from other sources

Source	Estimates	Year	Method
China Household Finance Survey by SWUFE	From 18.4% to 21.4%	2011, 2013, 2015 and 2017	Based on over 40,000 households in 364 districts or county-level cities in 29 provinces
China International Capital Corporation	From 17.7% to 18.3%	2010, 2013	Based on the aggregated data from the National Population Census in 2010
Wu, Gyourko & Deng (2016)	From 5.2% to 7%	2009, 2013	Based on Urban Household Survey by NBSC (9 provinces between 2002 and 2009; annual sample size: 50,000 households)
Glaeser et al. (2017)	20% in Tier 1 cities 13% in lower-tier cities	2012	Based on the Chinese Urban Household Survey (UHS)

⇒ Large discrepancies!

Vacancy rate — *cont.*

- The China Household Finance Survey in 2017 by SWUFE:
 - Over 40,000 households in 364 districts or county-level cities in 29 provinces
 - Home vacancy rates in China's urban areas were **18.4%**, **19.5%**, **20.6%** and **21.4%** in 2011, 2013, 2015 and 2017 respectively

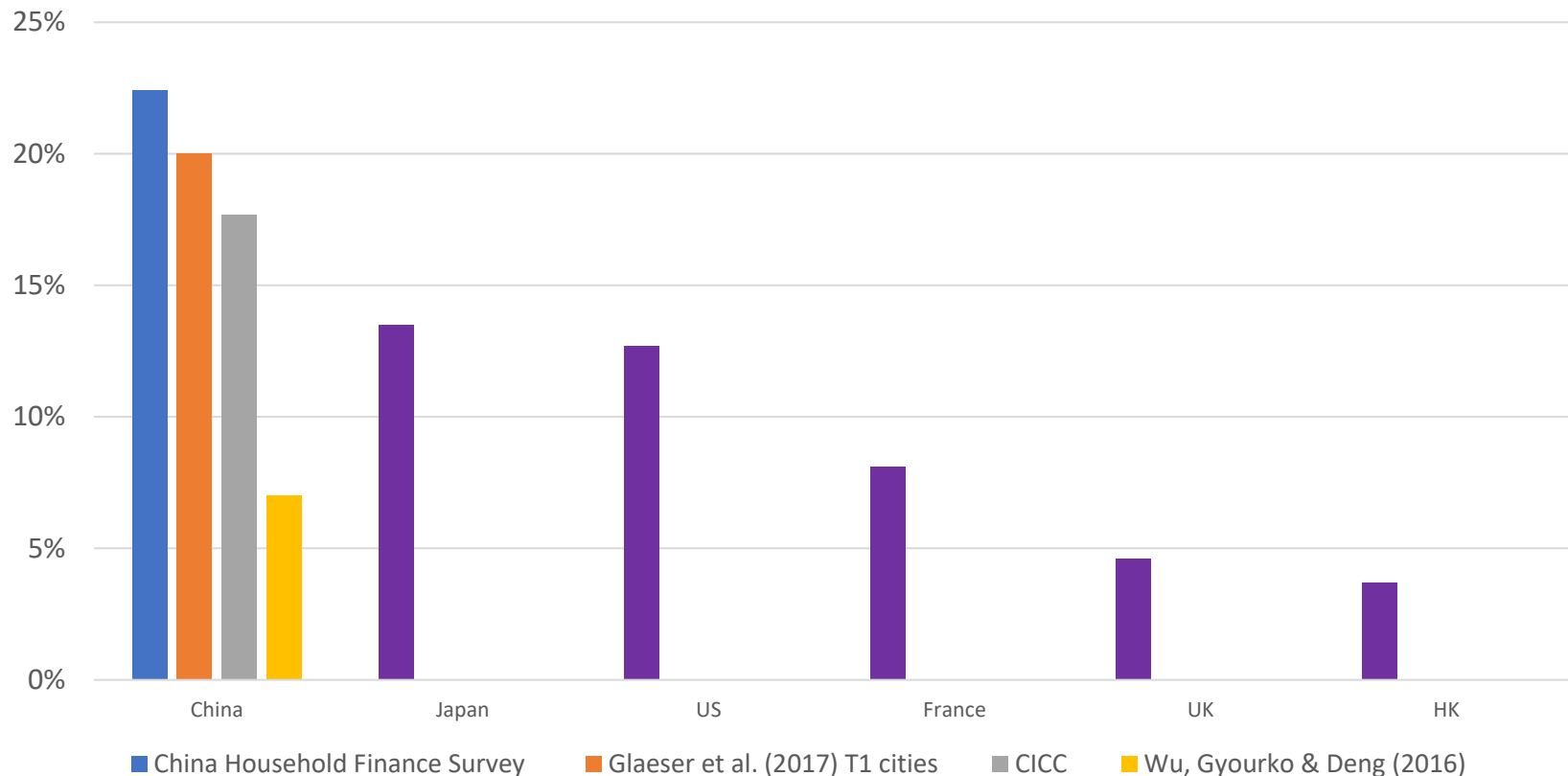
Bloomberg

Economics

**A Fifth of China's Homes Are Empty.
That's 50 Million Apartments**

... and very high in international comparison...

- Countries with inelastic long run supply of housing tend to have lower vacancy rates



Note: Japan in 2013, UK in 2014, US, France and HK in 2017

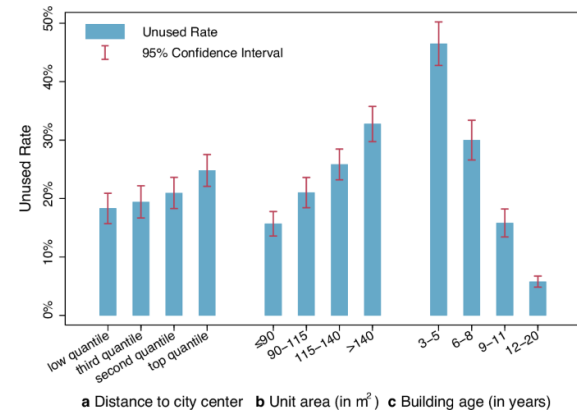
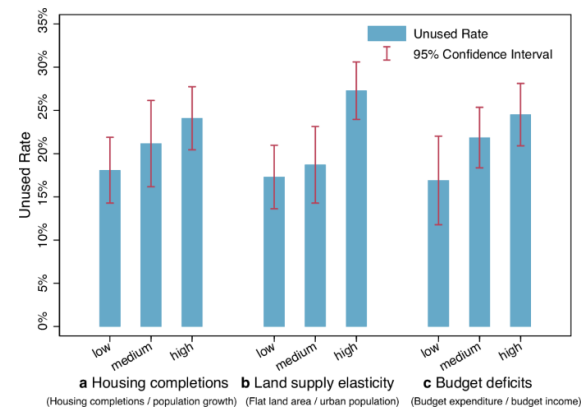
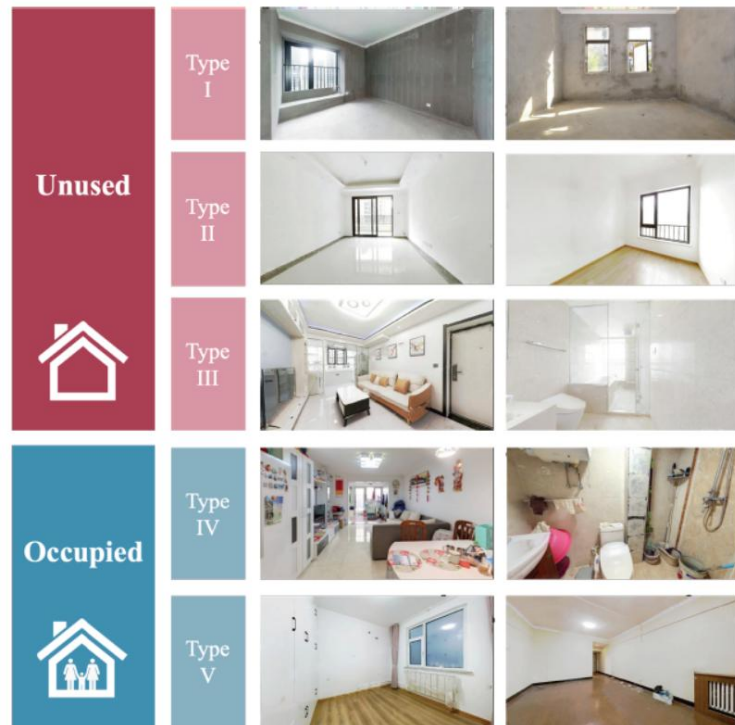
Source: H.K's Rating and Valuation Department, Japan's statistics bureau, China Household Finance Survey, U.S. Housing Vacancy Survey

<https://sdw.ecb.europa.eu/browse.do?node=9689360>

<https://www.bloomberg.com/news/articles/2018-11-08/a-fifth-of-china-s-homes-are-empty-that-s-50-million-apartments>

Unused housing in China

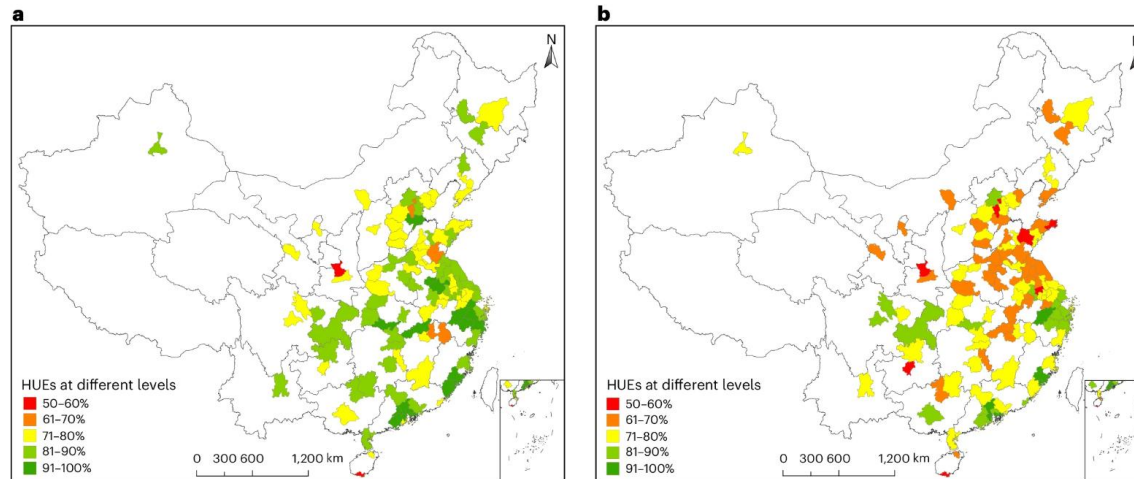
- Zheng *et al.* (2025) define “unused housing” as dwelling units that have been built and sold for at least two years but never occupied. They find that by early 2021, 17.4% of the housing stock built in China (2000-2020) remained unused
- The unused rate is particularly high in cities more likely to be associated with the oversupply of housing, especially in the majority of third-tier cities



Source: Zheng *et al.* (2025)

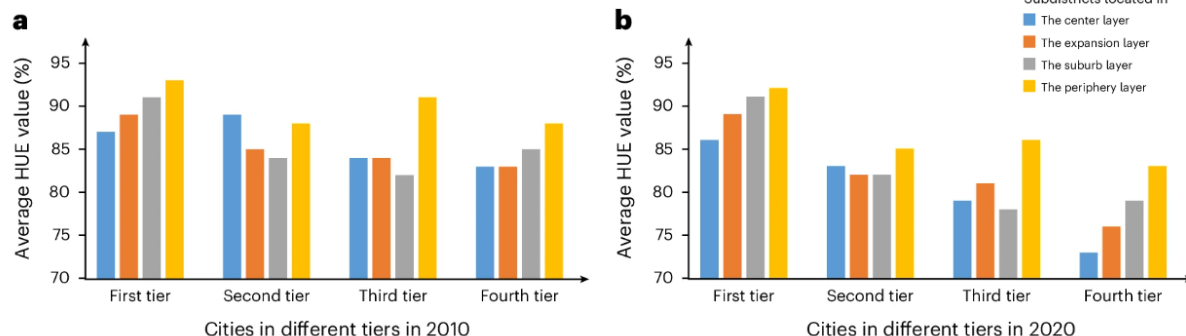
Housing utilization efficiency

- Shi et al. (2025) propose the concept of housing utilization efficiency (HUE), measured as the ratio between the actual population of a subdistrict and the population capacity of all housing units physically suitable for occupancy in the subdistrict



a, Regional differences of HUEs across China in 2010. **b**, Regional differences of HUEs across China in 2020.

The authors find that the overall HUE in China's highly urbanized areas decreased from 84% in 2010 to 78% in 2020



Source: Shi et al. (2025)

Possible causes for high vacancy rate in China

⇒ How to explain the comparatively high vacancy rate in China?

Discuss...

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Impact of fiscal stimulus program in 2008

- In 2008, China launched a massive fiscal stimulus program (4 trillion RMB ~ 12.5% of China's GDP) as an attempt to minimize the impact of the global financial crisis
- The stimulus package was invested in areas such as housing, infrastructure, transportation, health, education, environment, disaster rebuilding, etc.
- The program was seen as a success: China's economic growth recovered from 6% during 4Q 2008 and 1Q 2009 to over 8% in Q2 2009 and over 10% in Q3 2009
- However, the stimulus package has been criticized for causing a surge in local government debt in China after 2009

Impact of fiscal stimulus program in 2008

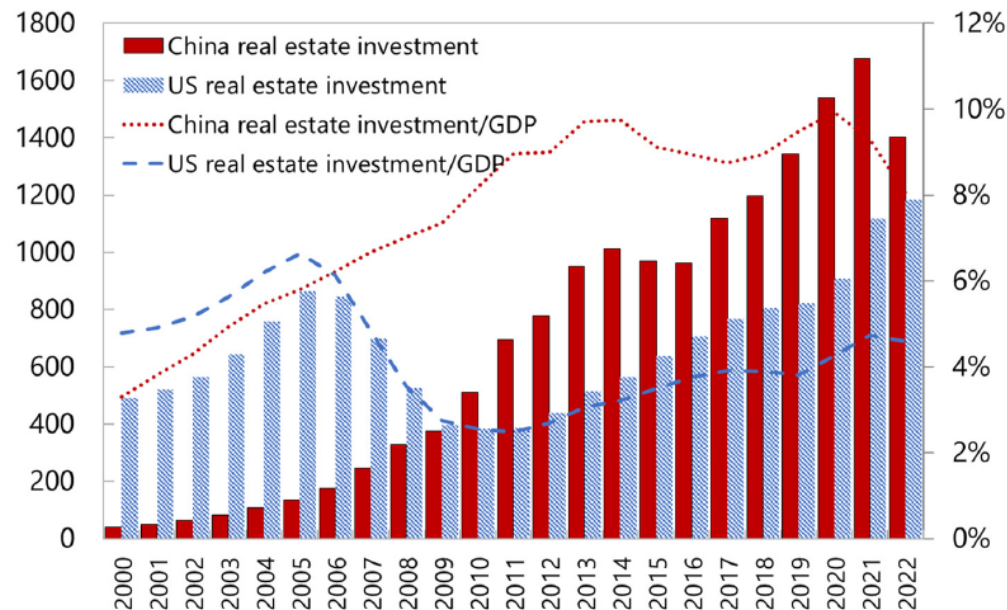
- Land and future land sale revenues serve as key collaterals for local governments to raise debt & finance infrastructure
- By 2013: Volume of outstanding local government debt reached 10.9 trillion RMB (~19.2% of China's GDP). 37% of local government's debt is collateralized by future land sale revenue (Liu and Xiong, 2018)
- Land revenue to repay the debt & LG's fiscal reliance on land revenue: one motivation for LG to implement restriction policies (instead of property tax) in China?

Real estate investment and GDP

- Real estate investment exceeds 10% of GDP in China in 2020 and 2021, higher than the US level before the Great Financial Crisis

Real Estate Investment in China and the United States

(Left: billion US dollars; Right: percent)



Sources: NBS, BEA and author calculations

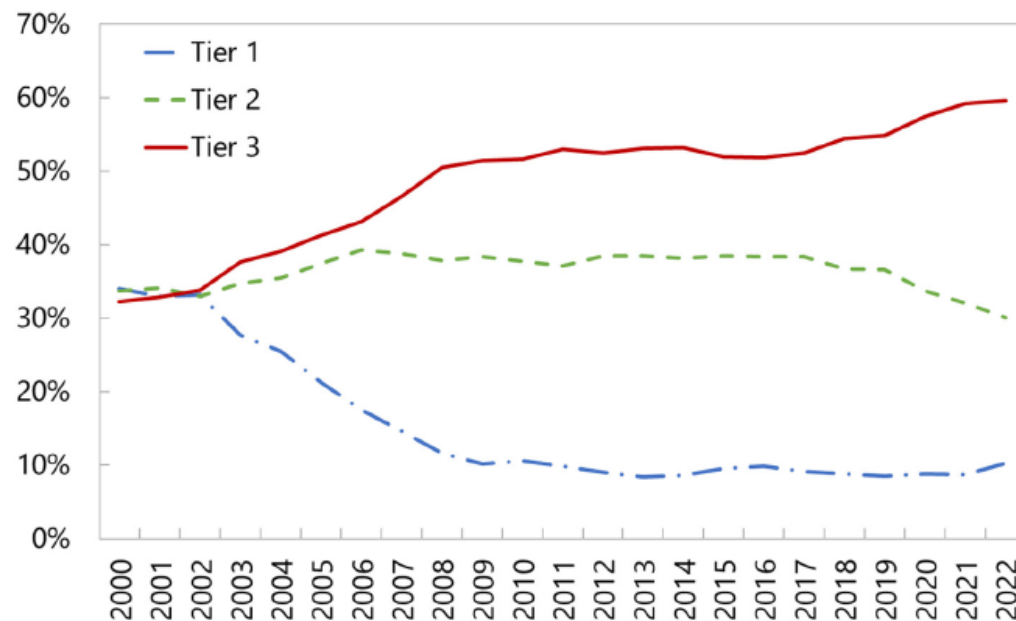
Fig. 2. Real Estate Investment in China and the United States.

Source: Rogoff and Yang (2024)

RE investment and local government debt

- Rogoff and Yang (2024) find that a 1 percentage increase in the RE investment ratio is associated with an approximately 3% increase in local governments' total debt scale

Share of Real Estate Investment by City Tier



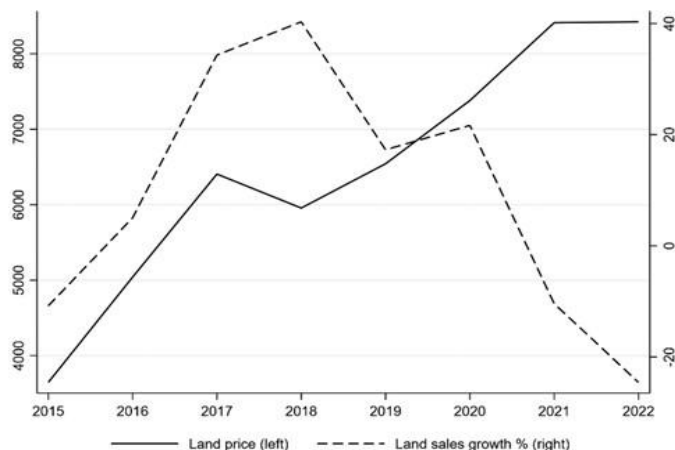
Sources: NBS, CEIC and author calculations

Fig. 5. Share of Real Estate Investment by City Tier.

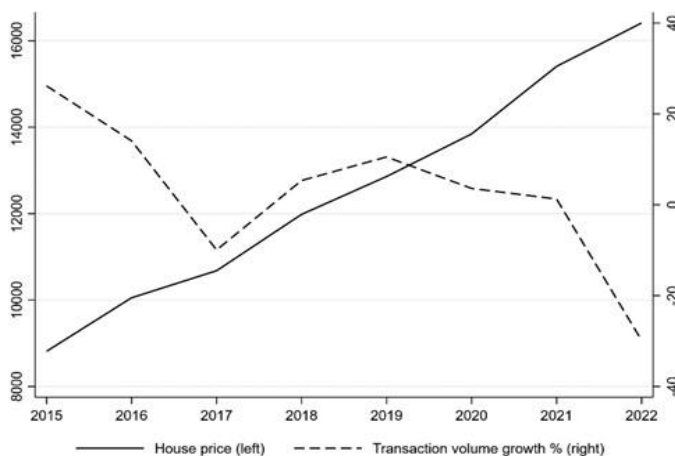
Source: Rogoff and Yang (2024)

Local governments and housing markets

A Land price and transaction volume (area) growth rate



B New housing price and transaction volume (area) growth rate



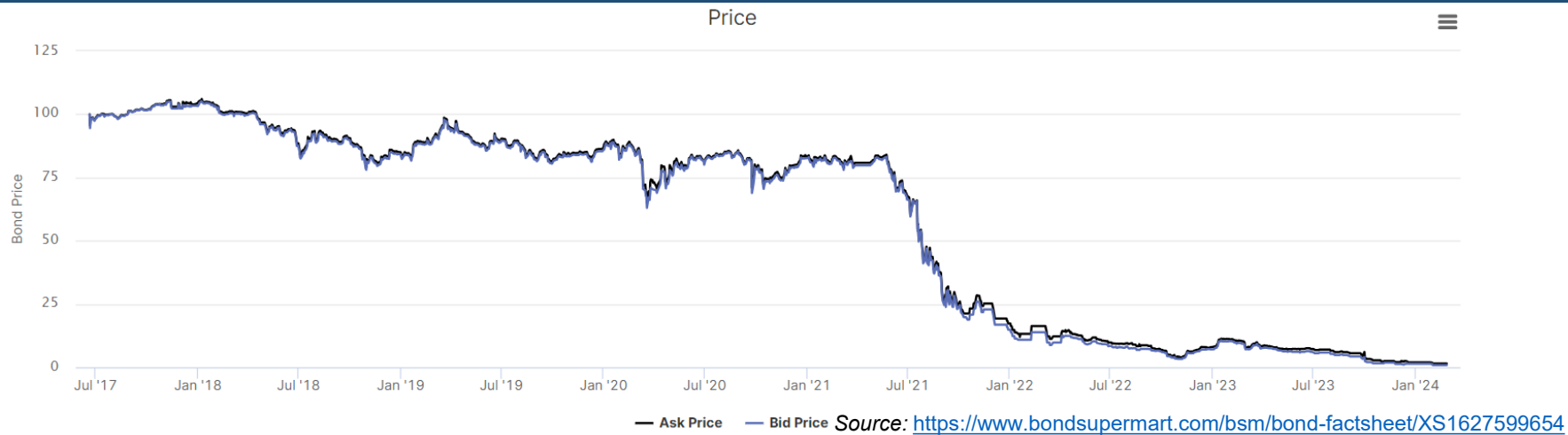
- Chang, Wang, and Xiong (2026) find that during the COVID pandemic (2020-2022), residential land and property prices surged even as transaction volumes plummeted
- The authors attribute this phenomenon to local governments' active price management through supply controls, land acquisitions by local government financing vehicles (LGFVs), and limits on new home sales permits

Source: Chang, Wang, and Xiong (2026)

Evergrande debt issue

- Evergrande is one of the largest RE developers in China
- In September 2021, the developer had 2 trillion RMB (310 billion USD) in liabilities
- In 2021, the company missed several debt payments and was downgraded by international ratings agencies:
 - Fitch downgraded Evergrande from B+ to CCC+
 - Moody's downgraded Evergrande's rating from B2 to Caa1
 - S&P Global Ratings downgraded Evergrande from B- to CCC
 - Fitch downgraded Evergrande further from CCC+ to CC
- The company finally defaulted on an offshore bond at the beginning of December 2021
- On 29 January 2024, a Hong Kong-based court ordered Evergrade to liquidate due to a lack of a viable restructuring plan

Evergrande bond & stock prices



Source: <https://www.google.com/finance/quote/EGRNQ:OTCMKTS?sa=X&ved=2ahUKEwj--TCtsuEAXWZ-AIHhcNXAIEQ3ecFegQlShAf&window=MAX>

Evergrande debt issue: potential causes

- Expansion and diversification strategy:
 - Evergrande's land reserves alone were large enough to house 10 million people in 2020
 - Evergrande had pursued an aggressive expansion, including ventures in electric vehicles, theme parks, energy, etc.
- High financial leverage through wealth management products (WMP): total WMP liabilities stood at 40 billion RMB in September 2021
- Policies to control rising HPs:
 - The "three red lines": government regulates the leverage taken on by developers, limiting their borrowing based on the following metrics: debt-to-cash, debt-to-equity, and debt-to-assets
 - Sale price caps in some cities
- COVID, aging population, and the expected future growth rate (Gordon growth model), especially for the lower-Tier cities

A political economy perspective

- Li and Zhou (2005) employ turnover data of top provincial leaders and economic outcome data in China between 1979 and 1995
- Key findings:
 - The likelihood of promotion of provincial leaders increases with their economic performance
 - The likelihood of being 'sidelined' decreases with their economic performance
- Interpreted by authors as evidence that China's central government utilizes personnel control to induce desirable economic outcomes: politician tournament theory

Contents

1. Some general stylized facts on Chinese cities
2. China's housing affordability issues
 - a) The problem
 - b) Potential causes
3. Implemented policies and policy evaluation
4. Other housing related problems (*if there is time*)
 - a) Overbuilding and ghost towns: Facts and potential causes
 - b) Local government debt: Facts and potential causes
- 5. Conclusions**

Conclusions

- Real estate industry is crucial for China's economy. In 2017, housing sales totaled 13.37 trillion RMB, equivalent to 16.4% of China's GDP (Liu and Xiong, 2018)
- According to estimates from World Bank, China is the second largest economy in the world in terms of GDP (14.3 trillion USD ~ around 16% of World's GDP in 2019). The up and down of China's housing market also affect the global economy
- Government implemented different housing policies to address the affordability crisis: a comparative study of housing policies in the developed countries

Takeaways

- Many interesting research questions e.g. for REEF dissertations
 - Policy evaluation (whether/to what extent housing policies succeed in China?)
 - Why housing prices are so high in China given the high vacancy rate and price-to-income ratio?
- Data sources
 - National Bureau of Statistics, Department of land and resources
 - China City Statistical Yearbook, China Real Estate Statistics Yearbook
 - Wharton/Tsinghua Chinese Residential Land Price Indexes (CRLPI)
 - Brokers such as Lianjia, Fang.com, Anjuke
 - China Household Finance Survey by SWUFE (online application)
 - Orbis, Wind, Bloomberg
 - [Housing affordability panel data](#) (Li, Qin, and Wu 2020)

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